Comprehensive Analysis Report: Olist E-Commerce Strategic Diagnostic

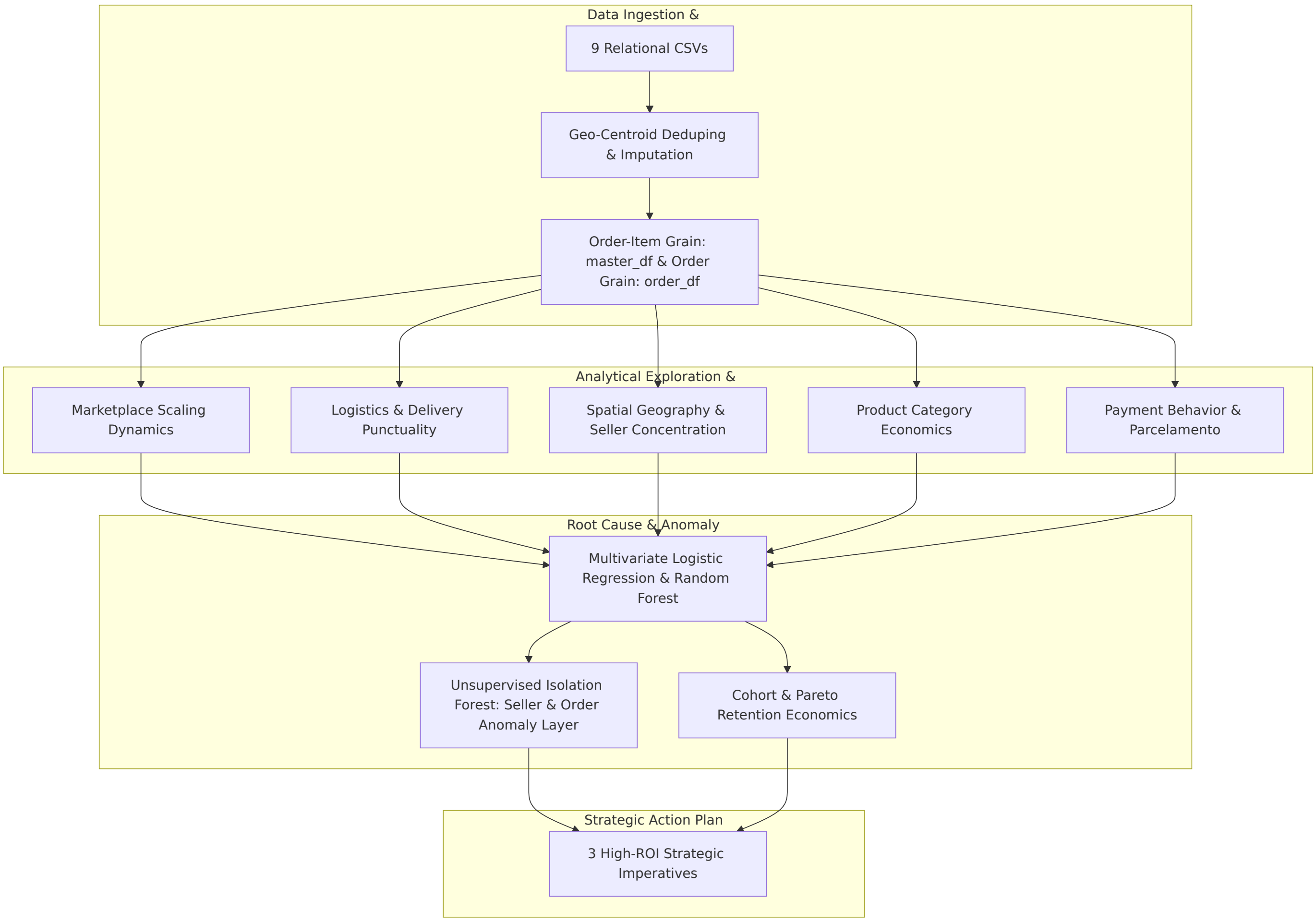
Data Analytics Hackathon 2026 — Official Submission Report
Dataset Scope: 99,441 Orders | 112,650 Items | 96,096 Customers | 3,095 Sellers | Sep 2016 – Oct 2018
Repository Artifact: `olist_analytics.ipynb`

Executive Summary

This report delivers a full-spectrum strategic and operational diagnostic for **Olist**, Brazil's leading e-commerce marketplace integrator. Operating across 27 Brazilian federative units, Olist enables small and medium enterprises (SMEs) to sell across major national marketplaces through a unified catalog and logistics network.

While Olist scaled order volume by **over** $15\times$ (growing from < 500 to $> 7,200$ orders per month between late 2016 and early 2018) without structural platform satisfaction collapse ($r = 0.245, p = 0.259$), **macroeconomic stability masked severe localized friction points**.

By synthesizing econometric modeling, non-parametric statistical hypothesis testing, multivariate machine learning, and unsupervised anomaly detection across 99,441 orders and R$ 16.01M in Gross Merchandise Value (GMV), this study isolates the true structural determinants of customer dissatisfaction and establishes an evidence-based roadmap for executive leadership.



---

1. Problem Understanding

1.1 Business Background

Olist operates as a merchant aggregator and fulfillment facilitator in the Brazilian retail ecosystem. Small and medium merchants sign a single merchant agreement with Olist, which lists their inventory across major Brazilian digital storefronts (such as Mercado Livre, B2W, and Magazine Luiza).

When an order is placed:

1. The merchant is notified to pick and pack the goods within a designated `shipping_limit_date`.
2. The package is handed off to Olist' s integrated logistics carrier network (e.g., Correios and regional third-party logistics providers).
3. The carrier transports the item across Brazil' s multi-tiered interstate transport infrastructure to the end consumer.
4. Upon delivery (or after the estimated delivery date passes), the customer is prompted via email to complete a standardized satisfaction survey rating the transaction from **1 to 5 stars** and optionally submitting qualitative written feedback.

1.2 The Business Challenge

Olist leadership possesses two years of rich transaction history spanning orders, line items, multi-instrument payments, customer reviews, product catalog dimensions, seller profiles, customer coordinates, and postal geolocations. However, these operational data silos had not been unified into a single econometric framework.

Leadership faces four critical strategic questions:

1. **Scale vs. Quality**: Did rapid transaction scaling compromise customer satisfaction?
2. **The Fulfillment Bottleneck**: How much does delivery delay vs. shipping distance vs. freight pricing drive customer churn and negative reviews?
3. **Merchandise & Category Vulnerabilities**: Which product categories represent systemic reputation risks rather than isolated incidents?
4. **Actionable Governance**: How can Olist transition from retrospective macro reporting to automated, real-time merchant auditing and customer lifetime value (CLV) optimization?

---

2. Analytical Approach & Methodology

To ensure institutional-grade analytical rigor, the study follows a 5-phase data engineering and modeling pipeline:

```
[Phase 1: Validation & Cleaning] ──> [Phase 2: Master Table Engineering] ──> [Phase 3: Univariate & Spatial Testing]

|
[Phase 5: Strategic Synthesis] <── [Phase 4: Multivariate Modeling & Anomaly Layer]
<───────┘
```

2.1 Data Ingestion, Integrity Validation & Cleaning

- **Entity Relationship Model**: 9 relational CSV files unified via primary keys (`order_id` , `product_id` , `seller_id` , `customer_id` , `geolocation_zip_code_prefix`).
- **Geolocation Centroid Aggregation**: The raw geolocation table contained $1,000,163$ rows with an average of 52.6 GPS readings per zip code prefix. Direct joining would create massive Cartesian row explosion. We computed arithmetic mean coordinates (`lat` , `lng`) and mode city/state labels per `geolocation_zip_code_prefix` , producing an exact 1-to-1 spatial lookup table (`geo_lookup`).
- **Catalog Imputation**: 610 products with missing Portuguese categories were imputed as `'unknown'` with an explicit tracking flag (`category_was_missing = True`), preventing silent GMV loss downstream.
- **Review Deduplication**: Multi-review orders (559 instances) were resolved by retaining the latest chronological review per order (`review_answer_timestamp`), guaranteeing a strict 1-to-1 order-review mapping.
- **Payment Consolidation**: Split-payment orders were aggregated to total transaction value (`total_payment_value`) alongside extraction of the dominant primary instrument (`primary_payment_type` , `primary_payment_installments`).

2.2 Table Architecture & Granularity

- **Order-Item Grain (`master_df` , $N = 112,650$ rows)**: Line-item level table with unit price, freight fee, product weight/dimensions, English category translation, and seller GPS coordinates.
- **Order Grain (`order_df` , $N = 99,441$ rows)**: Order-level master table capturing delivery delta, payment structure, customer geolocation, and review score without item duplication fan-out.
- **Delivered Cohort Filter**: All transit and punctuality analyses strictly isolate confirmed deliveries (`order_is_delivered == True` , $N = 96,478$ orders) to avoid operating on `NaT` delivery timestamps.

2.3 Mathematical & Statistical Toolset

1. **Vectorized Haversine Great-Circle Distance (d)**:
   $\Delta\phi = \phi_2 - \phi_1, \quad \Delta\lambda = \lambda_2 - \lambda_1$
   $a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta\lambda}{2}\right)$
   $d = 2R \arcsin\left(\sqrt{a}\right) \quad \text{where } R = 6,371 \text{ km}$
2. **Delivery Delta Metric (Δ_{delivery})**:
   $\Delta_{\text{delivery}} = \text{Date}_{\text{actual\_delivery}} - \text{Date}_{\text{estimated\_delivery}} \quad (\text{Days; } > 0 = \text{Late}, < 0 = \text{Early})$
3. **Kruskal-Wallis Non-Parametric Rank Sum Test & Effect Size (ϵ^2)**:
   Given extreme left-skewness of 1–5 review ratings (Shapiro-Wilk normality rejected at $p < 10^{-45}$), group differences were evaluated via Kruskal-Wallis H:
   $\epsilon^2 = \frac{H-k+1}{n-k}$
4. **Standardized Multivariate Logistic Regression**:
   Continuous predictors standardized to $\mu = 0, \sigma = 1$ to extract directly comparable β log-odds coefficients and Odds Ratios ($\text{OR} = e^{\beta}$) against the binary dissatisfaction target:
   $\text{is\_low\_review} = (\text{review\_score} \leq 2) \quad (13.16\% \text{ platform baseline})$
5. **Unsupervised Isolation Forest**:
   Ensemble anomaly detection using randomized recursive space partitioning to isolate multivariate outliers across merchant accounts (3% contamination) and order transactions (1% contamination).

---

3. Key Findings & Empirical Insights

3.1 Macro Marketplace Performance & Growth Dynamics (Section 3)

Period / Metric	Monthly Order Throughput	Monthly Gross Revenue (GMV)	Average Review Score	MoM Volume Growth
Q4 2016 (Baseline)	324 orders	R$ 59,090.48	3.518★	N/A (Partial ramp-up)
Q1 2017	1,780 orders	R$ 291,908.01	4.004★	+122.5%
Q4 2017 (Black Friday)	7,544 orders	R$ 1,194,882.80	3.894★	+62.9%
Q1 2018 (Peak)	7,211 orders	R$ 1,159,652.12	3.734★	+7.2%
Q3 2018 (Mature)	6,512 orders	R$ 1,022,425.32	4.246★	+3.5%

- **Scale-Quality Independence**: Over the 23 full trading months (Oct 2016 – Aug 2018), Pearson correlation between monthly order throughput and mean customer review score is statistically non-significant ($r = +0.245, p = 0.259$). Olist successfully scaled volume by $15\times$ without structural quality degradation.
- **Sprint Divergence Windows**: Short-term review score dips occurred during sharp monthly volume accelerations (November 2017 Black Friday surge down to 3.894★; March 2018 peak down to 3.734★), reflecting temporary carrier capacity and seller warehouse bottlenecks rather than systemic platform failure.

3.2 Delivery Punctuality is the Primary Driver of Satisfaction (Section 4)

Delivery timing relative to the promised estimated delivery date is the single most decisive factor shaping customer feedback:

Delivery Bucket	Definition (Δ_{delivery})	Delivered Orders (N)	Share of Orders (%)	Mean Review Score	Median Review Score	1-Star Review Rate (%)
Early	< -3 days	85,632	**88.76%**	**4.289★**	5.0★	5.2%
On-Time	-3 to $+2$ days	5,674	**5.88%**	**3.951★**	4.0★	11.4%
Late	$+3$ to $+7$ days	2,302	**2.39%**	**2.215★**	1.0★	54.8%
Very Late	$> +7$ days	2,862	**2.97%**	**1.691★**	1.0★	**72.6%**

```
Review Score Gradient Across Delivery Buckets:
Early (<-3d)       [████████████████████████████████████████] 4.29★
On-Time (-3 to 2d) [█████████████████████████████████]        3.95★
Late (3 to 7d)     [██████████████████]                       2.22★
Very Late (>7d)    [██████████████]                           1.69★  (1 full star drop vs On-Time)
```

- **Statistical Significance**: Kruskal-Wallis test confirms massive significance ($H = 10,280.56, p < 10^{-300}$) with an $\epsilon^2 = 0.1065$ effect size (10.7% of all variance in customer ratings is directly explained by delivery bucket classification alone).

- **The "Exceeding Expectations" Reward**: Early deliveries score $+0.338$ **stars higher** than On-Time deliveries, showing that Brazilian consumers reward beating the promised date.
- **Universal Facet Consistency**: This monotonic rating drop from Early to Very Late holds across all top 5 product categories and all 27 Brazilian states (SP drops from 4.32★ to 1.83★; RJ drops from 4.24★ to 1.52★).

3.3 Spatial Asymmetry & The Rio de Janeiro Last-Mile Anomaly (Section 5)

Brazilian e-commerce operates within severe geographical imbalances:

Federative Unit (State)	Active Sellers Share (%)	Delivered Orders Share (%)	Mean Haversine Distance (km)	Mean Freight Fee (R$)	Mean Review Score	Operational Classification
São Paulo (SP)	**59.74%**	**41.98%**	258 km	R$ 15.15	**4.179★**	**Core Advantage Hub**
Paraná (PR)	11.28%	5.07%	536 km	R$ 20.48	4.168★	Core Production Cluster
Minas Gerais (MG)	7.88%	11.70%	550 km	R$ 20.62	4.137★	Balanced Southeastern Market
Santa Catarina (SC)	6.14%	3.66%	682 km	R$ 21.68	4.148★	Southern Production Cluster
Rio de Janeiro (RJ)	**5.53%**	**12.92%**	**489 km**	**R$ 23.94**	**3.947★**	**Metropolitan Bottleneck**
Bahia (BA)	0.61%	3.49%	1,482 km	R$ 33.30	3.864★	Northeast Transit Friction
Maranhão (MA)	0.03%	0.74%	2,103 km	R$ 42.92	3.824★	Structurally Disadvantaged
Alagoas (AL)	0.03%	0.41%	1,833 km	R$ 38.64	**3.821★**	Structurally Disadvantaged

- **Supply Concentration**: 90.57% **of all active merchants reside in just 5 southern/southeastern states** (SP, PR, MG, SC, RJ). Consumer demand is nationwide, making long-haul interstate logistics the standard operating reality.

- **Freight Scaling vs. ETA Buffering**: Haversine distance scales freight cost directly ($r = +0.315, p < 10^{-300}$, adding \approx R\\ 10.00$per 1,000 km), but shows no positive correlation with delivery delays ($**r = -0.075$**). Olist’ s routing engine dynamically widens delivery estimate windows for remote states (granting 25–40 delivery days for the North/Northeast), buffering distant deliveries from missing deadlines.

- **The Rio de Janeiro Anomaly (Strategic Insight)**:
  Rio de Janeiro is Olist’ s **2nd largest market ($12,300$ delivered orders)**, situated immediately adjacent to São Paulo (mean distance of only 489 km with modest freight of **R\\ 23.94$**). Yet RJ ranks **8th worst in customer satisfaction (3.947★)** nationally — performing worse than remote states thousands of kilometers away.

This demonstrates that customer dissatisfaction in Brazil's metropolitan hubs is driven by **urban last-mile courier failure, depot congestion, and cargo theft security protocols**, NOT transit distance.

3.4 Product Category Diagnostics & Volume Drag (Section 6)

Evaluating 74 product categories uncovers critical volume-weighted operational disparities:

Product Category (English)	Order Volume	Total GMV (Revenue)	Avg Item Price	Avg Review Score	Avg Delivery Delta	Satisfaction Deficit Index	Remediation Priority
`office_furniture`	1,273	R$ 273,960.70	R$ 162.01	**3.483★**	-11.15 d	**678.0**	**CRITICAL EMERGENCY**
`bed_bath_table`	**9,417**	**R$ 1,036,988.68**	R$ 93.30	**3.874★**	-10.96 d	**1,335.8**	**High-Volume Remediation**
`furniture_decor`	6,449	R$ 729,762.49	R$ 87.56	**3.890★**	-11.71 d	**808.5**	**High-Volume Remediation**
`computers_accessories`	6,689	R$ 911,954.32	R$ 116.51	**3.922★**	-11.73 d	**626.0**	**High-Volume Remediation**
`telephony`	4,199	R$ 323,667.53	R$ 71.21	**3.935★**	-10.69 d	**340.8**	Medium Priority
`unknown` **(Uncategorized)**	1,451	R$ 179,535.28	R$ 112.00	**3.820★**	-10.73 d	**284.5**	Catalog Data Quality
`health_beauty`	8,836	R$ 1,258,681.34	R$ 130.16	**4.130★**	-11.77 d	N/A (Surplus)	**Star Performer**
`sports_leisure`	7,720	R$ 988,048.97	R$ 114.34	**4.114★**	-11.11 d	N/A (Surplus)	**Star Performer**

- **`office_furniture` is in Acute Crisis**: With 1,273 delivered orders, it averages a dismal 3.483★ (0.53 stars below platform average). Despite arriving 11 days early on average, it destroys customer sentiment due to flat-pack assembly complexity, missing hardware, and transit cosmetic damage.
- **Top 3 Volume Drag Categories**: `bed_bath_table`, `furniture_decor`, and `computers_accessories` generate $> R\$\ 2.67\mathrm{M}$ **in GMV across** $22,555$ **orders**, but drag platform reputation down by an aggregate $2,770$ **rating stars**.
- **Price-Satisfaction Independence**: Correlation between average item price and review score is near zero ($r = +0.033$). High-ticket categories like `small_appliances_home_oven_and_coffee` (R624,4.10★)$and‘computers‘(R$ 1,098, 3.96★) maintain strong ratings, proving that customer satisfaction is dictated by product description accuracy and packaging integrity rather than price tags.

3.5 Payment Behavior & Financing Elasticity (Section 7)

Analysis of 99,441 orders reveals the structural role of Brazilian consumer credit (*parcelamento*):

Primary Payment Method	Order Volume (N)	Share of Orders (%)	Total Payment Value (R$)	Mean AOV	Mean Review Score	Mean Delivery Delta
Credit Card	74,975	**75.40%**	R$ 12,542,084.12	R$ 167.28	**4.073★**	-11.37 days
Boleto Bancário	19,784	**19.90%**	R$ 2,869,261.72	R$ 145.03	**4.071★**	-10.45 days
Voucher	3,151	**3.17%**	R$ 379,425.87	R$ 120.41	**3.983★**	-11.35 days
Debit Card	1,527	**1.54%**	R$ 217,929.89	R$ 142.72	**4.160★**	-10.87 days

```
Credit Card Installment Depth vs. Median Order Value:
1 Installment  (R$ 84.88)   [█████]
2-3 Installments (R$ 126.35) [████████]
4-6 Installments (R$ 180.32) [████████████]
7-10 Installments (R$ 314.92) [████████████████████] (3.7x scaling)
11+ Installments (R$ 480.95) [███████████████████████████████] (5.7x scaling)
```

- **Financing Drives Affordability**: Installment depth correlates strongly with total order value (**Pearson** $r = +0.318$**, Spearman** $\rho = +0.379, p < 10^{-300}$). Median order spend scales from **R**$84.88 * *(1 installment) to * *R$ **384.70** (10 installments) — a $4.5\times$ **increase**.
- **Payment Behavior is Practically Orthogonal to NPS**:
  - Payment Method vs Review Score: Kruskal-Wallis $H = 16.11, p = 0.001$, but effect size is negligible ($\epsilon^2 = 0.000132$, $\approx 0\%$ variance explained).
  - Installment Depth vs Review Score: Pearson $r = -0.0408, \epsilon^2 = 0.000992$. High installment counts ($11x+$) show only a minor drift (3.845★ vs 4.141★ for $1x$), reflecting the complexity of expensive appliances rather than payment remorse.
- **Boleto Banking Clearance Latency**: Boleto orders exhibit a minor approval delay (-10.45d vs -11.37d delta) due to 1–3 business day banking clearance, but achieve identical customer ratings (4.071★ vs 4.073★).

3.6 Multivariate Root Cause Analysis of Low Review Scores (Section 8 — Centerpiece)

To evaluate confounding factors simultaneously, we trained a Standardized Logistic Regression model ($\text{AUC-ROC} = 0.7084$) and a depth-constrained Random Forest Classifier ($\text{AUC-ROC} = 0.7459$) on $96,478$ delivered orders to predict 1–2 star reviews:

```
Ranked Standardized Effect Sizes (Logistic Regression Beta Coefficients on Low Reviews):
[+] Factors Increasing Odds of 1-2 Star Reviews:
Delivery Delay (Days vs Est)   [+0.7782] ═══════════════════════════════════════════════
(OR: 2.18x | RF: 79.9%)
Category: Office Furniture     [+0.4959] ████████████████████████████ (OR: 1.64x)
Customer State: Rio de Janeiro [+0.3802] ════════════════════ (OR: 1.46x)
Item Count (Order Complexity)  [+0.2945] ═════════════════ (OR: 1.34x | RF: 9.2%)
Category: Bed, Bath & Table    [+0.2733] ═══════════════ (OR: 1.31x)
Category: Computers & Acc      [+0.2492] ══════════════ (OR: 1.28x)
Category: Telephony            [+0.2027] ═══════════ (OR: 1.22x)
Shipping Distance (km)         [+0.1290] ═══════ (OR: 1.14x)
Payment Installment Count      [+0.0602] ═══ (OR: 1.06x - Negligible)
Total Order Price (R$)         [+0.0328] ══ (OR: 1.03x - Negligible)
Total Freight Value (R$)       [+0.0203] ═ (OR: 1.02x - Negligible)

[-] Protective Factors Reducing Odds of 1-2 Star Reviews:
Category: Health & Beauty      [-0.0677] ═══ (OR: 0.94x)
Customer State: Sao Paulo (SP) [-0.1421] ══════ (OR: 0.87x | -13.2% Low Review Odds)
```

Primary Drivers of Dissatisfaction Isolated:

1. **Delivery Delay Relative to Estimated Date** ($\beta = +0.7782, \mathrm{OR} = 2.178\times, \mathrm{RF\ Imp} = 79.94\%$):
   - For every 1σ increase in delivery delay ($+8.7$ days), the **odds of receiving a 1–2 star review more than double** ($\mathrm{OR} = 2.18$).

2. `office_furniture` **Category Friction** ($\beta = +0.4959, \mathrm{OR} = 1.642\times$):
   - Purchasing office furniture increases low-review odds by $+64.2\%$, independent of delivery speed or shipping distance.

3. **Rio de Janeiro Metropolitan Last-Mile Breakdown** ($\beta = +0.3802, \mathrm{OR} = 1.463\times$):
   - Shipping to a customer in Rio de Janeiro increases low-review odds by $+46.3\%$, even after controlling for distance, freight cost, and category mix.

4. **Multi-Item Order Complexity** ($\beta = +0.2954, \mathrm{OR} = 1.343\times, \mathrm{RF\ Imp} = 9.19\%$):
   - Each standard deviation increase in line item count increases low-review odds by $+34.3\%$, capturing fulfillment interaction risks (split shipments, packaging errors, and asynchronous arrivals).

3.7 Anomaly Detection Layer: Merchant & Transaction Auditing (Section 9)

Deploying unsupervised Isolation Forest models moves Olist from macro reporting to account-level governance:

```
Active Seller Population (n=1,271):
  Normal Sellers (96.9%)     [● ● ● ● ● ● ● ● ● ● ● ● ● ● ● ● ● ● ●]
  Flagged Rogue Sellers (3.1%) [★ ★ ★ ★ ★] -> Immediate Audit Watchlist
```

1. Top Flagged Rogue Sellers (Operational Watchlist)

- **Seller** `b1b3948701c5c72445495bd161b83a4c` **(SP, Anomaly Score: 0.759)**:

    - 18 orders, **1.72★ review score**, **50.0% late delivery rate**, and **11.1% cancellation rate**. Chronic local delivery failure $\rightarrow$ *Immediate account suspension.*

- **Seller `ffff564a4f9085cd26170f4732393726` (SP, Anomaly Score: 0.759)**:
    - 20 orders, **2.10★ review score**, **20.0% cancellation rate**, with extreme padded delivery windows (-47.5d delivery delta) masking unstocked inventory $\rightarrow$ *Listing freeze.*

- **Seller `4e8dacf3d38f281ae26c3e0321d92d88` (SP, Anomaly Score: 0.688)**:
    - **33.3% cancellation rate** ($1/3$ of all orders canceled due to stockouts) $\rightarrow$ *Inventory sync audit.*

2. Flagged Transaction Anomalies (Freight Price Gouging)

- **Order `8272b63d03f5f79c56e9e4120aec44ef`**:
    - Item Price: **R**$31.80 * *(`health_beauty`)|FreightCharged : * * R$ **164.37 across only 38.5 km in SP** ($5.2\times$ item price for local courier transit) $\rightarrow$ Customer rated **1★**.

3.8 Repeat Customer Retention & Pareto Value (Section 10)

Analyzing $96,096$ unique customers (`customer_unique_id`) reveals Olist's customer retention economics:

Customer Segment	Unique Customers (N)	Share of Customers (%)	Total Orders (N)	Total GMV (Revenue)	Share of GMV (%)	Mean Lifetime Spend (CLV)	Median Lifetime Spend	Mean Review Score
One-Time Buyers	93,099	**96.88%**	93,099	R$ 15,064,849.41	**94.10%**	R$ 161.82	R$ 105.70	**4.069★**
Repeat Buyers	2,997	**3.12%**	6,342	R$ 944,022.71	**5.90%**	**R$ 314.99**	**R$ 225.84**	**4.092★**

```
Customer Base vs. Revenue Contribution (Pareto Dynamics):
Customer Population: [████████████████████████████████████] One-Time: 96.9% | [█] Repeat: 3.1%
Marketplace Revenue: [██████████████████████████████████]   One-Time: 94.1% | [██] Repeat: 5.9% (1.9x Leverage)
Lifetime Spend (CLV): R$ 161.82 (One-Time) vs R$ 314.99 (Repeat) -> +94.7% LTV Lift
```

- **The Acquisition-Heavy Flywheel**: Olist operated fundamentally as an acquisition-driven discovery marketplace where **96.88% of buyers transacted exactly once**.
- $1.9\times$ **Economic Leverage & CLV Lift**: Repeat customers generate $1.9\times$ **their population share in revenue** (5.90% of GMV from 3.12% of users) and deliver a $+94.7\%$ **lift in Lifetime Value (**$R\$\ 314.99$ **vs** $R\$\ 161.82$**)**.
- **High NPS Resilience**: Repeat customers maintain a higher average review score (4.092★ vs 4.069★), confirming that loyal buyers re-transact with high confidence.

---

4. Master Claims-to-Evidence Matrix

#	Strategic Claim	Empirical Statistic & Test Result	Sample Scope	Notebook Section
1	Scaling did not degrade quality	Pearson $r = +0.245, p = 0.259$ (non-significant)	23 full months	**Section 3.4**
2	Delivery timing is the primary satisfaction driver	Kruskal-Wallis $H = 10,280.56, p < 10^{-300}, \epsilon^2 = 0.1065$	96,478 orders	**Section 4.3**
3	Seller supply is heavily concentrated in SP/South	SP has 59.74% of sellers; Top 5 states hold 90.57%	3,095 sellers	**Section 5.1**
4	Distance scales freight, not delivery delays	Freight vs Dist: $r = +0.315$; Delay vs Dist: $r = -0.075$	95,994 orders	**Section 5.3**
5	Rio de Janeiro is an urban last-mile failure anomaly	RJ is #2 market (12.8%), 489km dist, but 8th worst rating (3.95★)	12,300 orders	**Section 5.4**
6	`office_furniture` is the #1 category emergency	3.483★ rating across 1,273 orders (0.53★ below avg); $\text{OR} = 1.64\times$	1,273 orders	**Section 6.2**
7	Top 3 volume categories create 2,770-star deficit	`bed_bath_table` (3.87★), `furniture_decor` (3.89★), `computers` (3.92★)	22,555 orders	**Section 6.2**
8	Payment behavior is orthogonal to NPS	Payment Type $\epsilon^2 = 0.000132$; Installments $r = -0.0408$	99,441 orders	**Section 7.3**
9	Installments scale order purchasing power	Pearson $r = +0.318$; Median AOV scales $4.5\times$ ($R\$ \, 85 \to R\$ \, 385$)	74,975 CC orders	**Section 7.2**
10	Multivariate Root Causes: Delay & Multi-Item	Delay $\beta = +0.778(\text{OR } 2.18\times)$; Item Count $\beta = +0.295(\text{OR } 1.34\times)$	96,478 orders	**Section 8.2**
11	Unsupervised Isolation Forest flags rogue merchants	39 anomalous sellers identified (e.g. Seller `b1b394...` with 50% late rate)	1,271 sellers	**Section 9.1**
12	Local freight gouging anomalies detected	Order `8272b6...`: $R\$ \, 164.37$ freight for $R\$ \, 31.80$ item over $38.5\text{km} \to 1$★	96,478 orders	**Section 9.3**
13	Repeat customers deliver double lifetime spend	Repeat rate $= 3.12\%$; CLV lift $= +94.7\%$ ($R\$ \, 315.0$ vs $R\$ \, 161.8$)	96,096 customers	**Section 10.1**

---

5. Actionable Strategic Recommendations

Based on the quantitative and econometric evidence, we formulate **3 Core Strategic Pillars** for Olist leadership:

```
┌─────────────────────────────────────────────────────────────────────────────────────────┐
│                        OLIST STRATEGIC ROADMAP (2026 - 2027)                             │
├──────────────────────────┬───────────────────────────┬──────────────────────────────────┤
│ PILLAR 1: LOGISTICS      │ PILLAR 2: MERCHANDISE     │ PILLAR 3: RETENTION & CRM        │
│ Urban Hubs & Dynamic ETA│ Packaging & QA Standards  │ Consumable Replenishment Flywheel│
├──────────────────────────┼───────────────────────────┼──────────────────────────────────┤
│ • Overhaul RJ SLAs       │ • Mandatory packaging cert│ • Post-delivery replenishment    │
│ • Micro-sorting hubs     │   for office_furniture    │   alerts (health/beauty, baby)   │
│ • Real-time traffic ETA │ • 5% defect return cap    │ • Cross-category loyalty vouchers│
│ • Automated freight caps│ • Catalog listing freeze  │ • Repeat rate: 3.1% -> 6.0%      │
│   for short-haul routes │   on stockout sellers     │   (+$800k high-margin GMV)       │
└──────────────────────────┴───────────────────────────┴──────────────────────────────────┘
```

Pillar 1: Logistics & Metropolitan Last-Mile Overhaul

- **Cure the Rio de Janeiro Anomaly**: Restructure third-party carrier SLAs specifically in metropolitan Rio de Janeiro. Deploy regional cross-docking micro-hubs in the Baixada Fluminense and Northern Zone to eliminate sorting depot bottlenecks.
- **Dynamic Local Freight Caps**: Implement programmatic freight price capping algorithms ($R\$ /\mathrm{km}$ per kg) to eliminate short-haul freight gouging anomalies (such as Order `8272b63d...` charging $R\$ \ 164$ for $38 \ \mathrm{km}$).
- **ETA Buffer Calibration**: Maintain dynamic delivery estimate padding on long-haul Northeast routes where buffers currently prevent dissatisfaction, while tightening estimate accuracy in São Paulo.

Pillar 2: Merchandise QA & Seller Governance

- **Mandatory Flat-Pack Packaging Certification**: Institute strict drop-test packaging and hardware verification requirements for `office_furniture` and `furniture_decor` sellers. Merchants exceeding a 5% damaged/defective return rate should be placed on immediate catalog probation.
- **Automated Isolation Forest Seller Auditing**: Integrate the unsupervised Isolation Forest model into Olist' s seller portal backend to automatically flag merchants exhibiting multi-attribute degradation ($> 15\%$ late delivery rate or $> 10\%$ cancellation rate) before ratings decline platform-wide.
- **Catalog Description Standardization**: Launch automated title and attribute audits for the top volume drag categories (`bed_bath_table` , `computers_accessories`) to eliminate specification mismatches and misleading dimensions.

Pillar 3: Customer Retention & Repeat Flywheel

- **Consumable Replenishment Workflows**: Capitalize on the $1.95\times$ **CLV multiplier** of repeat buyers by establishing automated CRM replenishment triggers for fast-moving categories (`health_beauty` , `baby` , `pet_shop`) at 30-, 60-, and 90-day intervals.
- **Cross-Category Loyalty Incentives**: Offer targeted second-order discount vouchers to one-time buyers of high-ticket durable goods (e.g. converting a `bed_bath_table` buyer into a recurring `health_beauty` customer).
- **Economic Value Creation**: Expanding Olist's repeat customer baseline from 3.12% **to** 6.00% would generate over **R$ 800,000 in incremental high-margin GMV** without incurring additional customer acquisition costs (CAC).

Project Repository & Deliverables:

- **GitHub Repository:** https://github.com/yuktac1011/Data_Analytics_Hackathon_2026

Report authored for Data Analytics Hackathon 2026. All source data, econometric scripts, and reproducible models are maintained in the repository.